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Asthma: pathophysiology, diagnosis and management

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Abstract

This article provides an overview of asthma in adults, including pathophysiology, risk factors and triggers. Assessment, diagnosis and pharmacological therapies are considered. The importance of working in partnership with the patient and encouraging supported self-care are highlighted.

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- Recognise risk factors for and triggers of asthma.
- Describe the key elements of assessment and diagnosis.
- Understand the pharmacological treatments for asthma.
- Explain the importance of partnership working between healthcare professionals and people with asthma.

Introduction

Asthma was first described by the ancient Greek physician Hippocrates and derived from the Greek word *asthmaino* meaning panting or gasping (Diamant *et al* 2007). Since ancient times, considerable advances have been made in understanding the genetics, epidemiology and pathophysiology of asthma, a condition that has increased in prevalence worldwide over the past 20 years (Rees 2010). It is estimated that 5.4 million people in the UK receive treatment for the condition, of which 1.1 million are children and 4.3 million are adults (Asthma UK 2004). International asthma guidelines have been in existence for over two decades and therapies of proven efficacy are available, yet controlling asthma can be difficult for many patients (Haughney *et al* 2008).

The disease significantly affects healthcare systems as well as the quality of life of patients and their families (Sarver and Murphy 2009). Poor asthma control results in increased hospital admissions and urgent care visits (Haughney *et al* 2008). In addition, persistent symptoms cause considerable morbidity and absence from work and school (Rees 2010).

Aims and intended learning outcomes

This article will focus on the pathophysiology, diagnosis and management of asthma in adults. After reading the article and completing the time out activities you should be able to:

- Identify the airways affected by asthma.
- Describe the pathophysiology of the disease.

Mortality rates associated with asthma have declined since the 1960s, but 1,131 deaths were reported in 2009 (Rees 2010). On average, three people per day or one person every eight hours dies from asthma in the UK (Asthma UK 2004). Improved care could avoid 75% of hospital admissions and up to 90% of deaths (Greener 2010). Therefore it is important that healthcare professionals who come into contact with patients with asthma have a good knowledge of the disease.

Complete time out activity 1

Defining asthma

There is no gold standard definition of asthma (British Thoracic Society (BTS) and Scottish Intercollegiate Guidelines Network (SIGN) 2011). However, the Global Strategy for Asthma Management and Prevention (Bateman *et al* 2008) describes the pathological, physiological and clinical features of the disease: 'Asthma is a chronic inflammatory disorder of the airways in which many cells and cellular elements play a role. The chronic inflammation is associated with airway hyper-responsiveness that leads to recurrent episodes of wheezing, breathlessness, chest tightness and coughing particularly at night or in the early morning. These episodes are usually associated with widespread, but variable, airflow obstruction within the lung that is often reversible either spontaneously or with treatment.'

Asthma can be attributable to atopy (Townshend *et al* 2007), but factors unrelated to atopy can also be important in the development of the disease (Anderson 2005).

Atopic asthma

Atopic asthma generally starts in childhood or adolescence and is associated with identifiable triggers that provoke wheezing. Atopic asthma is often associated with a family history of allergic diseases and features of atopy such as eczema and rhinitis (Diamant *et al* 2007, Townshend *et al* 2007). The disease commonly occurs as a result of an allergic response to specific allergens such as house dust mite, grass and tree pollen and dander from domestic pets (Ward *et al* 2010).

Exposure to an allergen in atopic individuals leads to the release of excessive quantities of Immunoglobulin E (IgE) from B lymphocytes. IgE binds to cells involved in inflammation, which stimulates the release of inflammatory mediators that cause bronchoconstriction and inflammation of the airways (Holgate

and Douglass 2010). Atmospheric pollution and maternal smoking in pregnancy can pre-dispose individuals to raised IgE levels and the development of asthma and airway hyper-responsiveness (Ward *et al* 2010).

Non-atopic asthma

Not all cases of asthma are attributable to atopy, therefore factors unrelated to atopic disease are also important (Anderson 2005). Some patients develop asthma in adult life often as a consequence of viral respiratory infections; this is often referred to as non-atopic asthma. This type of asthma can be more persistent with few obvious triggers other than infection (Diamant *et al* 2007, Rees 2010). Non-atopic asthma is IgE-independent (Ward *et al* 2010).

Pathophysiology

Airways affected by asthma

Asthma can affect the trachea, the bronchi and the bronchioles, which form part of the lower respiratory tract. The disease causes bronchoconstriction or abnormal narrowing of the airways as a result of epithelial damage, over-production of mucus, oedema, bronchospasm and muscle damage (Barnes 1996, Rees 2010).

Epithelial damage

In asthma, the epithelium (the layer of cells that line the airways) can become damaged and peel away. Epithelial shedding can contribute to airway hyper-responsiveness in several ways; these include loss of barrier function, which may allow penetration of allergens; loss of enzymes that break down inflammatory mediators; and exposure of sensory nerves, which may lead to reflex neural effects on the airway (Barnes 1996). Changes can also occur in the subepithelial layer, such as the laying down of collagen (Rees 2010).

Mucus hypersecretion

Asthma causes the mucus-secreting cells in the airways to multiply and the mucous glands to expand. Increased mucus secretion contributes to the formation of viscid mucous plugs that can occlude the airways (Ward *et al* 2010).

Oedema

The capillaries in the airway walls can dilate and may leak. The consequences of microvascular leakage include increased airway secretions, impaired mucociliary clearance and oedema,



TIME
OUT

1 Using an anatomy and physiology textbook, review the anatomy of the respiratory system. List the parts of the respiratory system that you think might be affected by asthma. How would you explain what asthma is to a patient?

which may contribute to airway narrowing and hyper-responsiveness (Barnes 1996).

Bronchospasm

Bronchospasm is also a feature of asthma. It describes the sharp contraction of bronchial smooth muscle, which narrows the airways.

Airway remodelling

With poorly controlled or undertreated asthma, changes in structural cells and tissues can occur in the lower respiratory tract that lead to remodelling of the airway, resulting in permanent fibrotic damage (Rees 2010).

Risk and trigger factors

The major risk factors for the development of asthma can be categorised as host or causal factors (Holgate and Douglass 2010). Host factors, such as a genetic predisposition to the disease, increase the risk of a person developing asthma. A familial link in asthma has been established; the chance of developing the disease by the age of 50 years is increased by ten times if there is a first-degree relative with asthma (Rees 2010). However, genetic susceptibility alone does not account for the development of the disease; it depends on environmental factors acting with a genetic predisposition (Rees 2010). Causal factors are environmental factors, for example indoor and outdoor allergens, air pollution and tobacco smoke, which influence susceptibility to the development of asthma in predisposed individuals (Holgate and Douglass 2010).

Trigger factors refer to the environmental influences that may instigate an asthma episode or cause symptoms to persist in an individual with the disease (Scullion 2005, Holgate and Douglass 2010). Examples of risk factors and triggers are shown in Box 1. **Complete time out activity 2**

Diagnosis

In the absence of a clear definition of asthma it can be difficult to diagnose the condition (BTS and SIGN 2011, McMurray 2010). Diagnosis requires clinical experience and judgement, because signs and symptoms of asthma can vary from patient to patient, and may change within the same patient at different times and under different circumstances (Sarver and Murphy 2009). Taking a clinical history is essential to allow a reasonably certain diagnosis of asthma, or an alternative diagnosis to be made (BTS and SIGN 2011).

Clinical diagnosis of asthma is often based on the presence of symptoms such as cough, wheeze, breathlessness and chest tightness (Holgate and Douglass 2010). However, these symptoms are not specific to asthma. A key indication that asthma may be the cause is symptom variability. Therefore, it is important to establish whether symptoms are intermittent. Additionally, it is essential to assess the severity of the symptoms by establishing how often they occur and whether they cause nocturnal waking or exercise limitation. It also important to identify whether symptoms are provoked

BOX 1

Risk factors and triggers for asthma

Host factors

- ▶ Genetic susceptibility.
- ▶ Atopy.

Causal factors

- ▶ Indoor allergens (house dust mite, animal allergens, cockroach allergen, fungi).
- ▶ Outdoor allergens (pollens, fungi).
- ▶ Active and passive smoking.
- ▶ Indoor and outdoor air pollution.
- ▶ Respiratory infections.
- ▶ Parasitic infections.
- ▶ Diet and drugs.
- ▶ Obesity.
- ▶ Socioeconomic status.
- ▶ Occupational sensitisers.
- ▶ Chemicals.

Trigger factors

- ▶ Allergens (house dust mite, pollens, cat allergen).
- ▶ Pollutants.
- ▶ Respiratory infections.
- ▶ Exercise.
- ▶ Weather changes.
- ▶ Sulphur dioxide.
- ▶ Emotions such as stress.
- ▶ Drugs (paracetamol and non-steroidal anti-inflammatory drugs).
- ▶ Smoke and dust.
- ▶ Foods and additives.
- ▶ Irritants such as household sprays and paint fumes.

(Scullion 2005, Holgate and Douglass 2010, Rees 2010)



2 Having read the previous section, summarise the key changes that take place in the airways as a result of asthma. List the main risk factors for development of the disease. What factors may trigger symptoms in an individual with asthma? A patient presents with suspected asthma. What steps need to be taken to confirm the diagnosis?

by specific triggers (Scullion 2005, Haldar and Pavord 2008).

Information about past medical history should include birth history of the patient, any history of asthma in childhood and details of any atopy. It is also important to establish any medications the patient is taking, including prescribed drugs, medications purchased over the counter and herbal or homeopathic remedies. Non-steroidal anti-inflammatory drugs, beta blockers and aspirin can exacerbate asthma (Currie *et al* 2009). Chronic cough caused by angiotensin converting enzyme (ACE) inhibitors may also mimic less well-controlled asthma (Currie *et al* 2009). The response to any treatment already trialled for respiratory symptoms, such as bronchodilators, should be established.

Family history, and specifically any history of asthma or atopy, should be explored.

Social history should include information about hobbies and pets to ascertain if these may be triggers. Common sources of inhaled allergens, which are important causal factors for asthma, include domestic animals such as cats, dogs, rabbits and guinea pigs, which release allergens in their saliva, urine, faeces and danders (Holgate and Douglass 2010).

Occupational asthma may account for as much as 9-15% of adult onset disease (BTS and SIGN 2011). Therefore asking patients about occupational exposure to airway irritants may reveal a trigger. The most frequently reported causative agents for occupational asthma include isocyanates, flour and grain dust, colophony and fluxes, latex, animals, aldehydes and wood dust (BTS and SIGN 2011). Individuals who are better when not at work or when on holiday should be referred and investigated for occupational asthma (Currie *et al* 2009). Workers reported to be at increased risk of developing asthma are highlighted in Box 2.

Although smoking and use of alcohol are not diagnostic markers for asthma, cigarette smoking is associated with persistent asthma and higher mortality if the individual experiences a near fatal exacerbation (Currie *et al* 2005). Similarly, alcohol misuse is associated with fatal or near fatal episodes (Currie *et al* 2005). Information about smoking behaviour and alcohol consumption can be used to inform health education for the patient. Since smokers and former smokers are at high risk for chronic obstructive pulmonary disease (COPD) (Kuebler *et al* 2008), information about smoking behaviour alerts the practitioner to potential differential diagnoses.

Assessment

As well as obtaining a clinical history, it is also important to obtain objective support for the diagnosis (BTS and SIGN 2011). Confirmation of asthma diagnosis depends on demonstrating airflow obstruction that varies over short periods of time (BTS and SIGN 2011). This can be achieved through spirometry (Scullion 2005) or peak expiratory flow (PEF) measurement.

Spirometry

Spirometry, which measures lung volumes and airflow (Scullion 2005, Bostock-Cox 2010), is the preferred test for the diagnosis and monitoring of patients with asthma because it allows clear identification of airflow obstruction (BTS and SIGN 2011). Spirometry can differentiate between restrictive and obstructive lung disease and measures the forced vital capacity (FVC) and the forced expiratory volume in the first second (FEV₁). Obstructive spirometry traces are seen in asthma and COPD (Bostock-Cox 2010). Tests of FVC and FEV₁ are conducted by asking the patient to perform a forced expiration after a maximum inspiration, and the highest of at least three reproducible measurements is recorded. The ratio of FEV₁ to FVC provides a useful measure of airway obstruction. Forced expiration normally results in FEV₁/FVC ratios of more than 70%. Ratios below 70% suggest airway obstruction and the lower the ratio, the more severe the obstruction (Holgate and Douglass 2010).

BOX 2

Workers at increased risk of developing asthma

- ▶ Bakers, food processors.
- ▶ Forestry workers.
- ▶ Chemical workers, plastics and rubber workers.
- ▶ Metal workers, welders.
- ▶ Textile workers.
- ▶ Electrical and electronic production workers.
- ▶ Storage workers.
- ▶ Farm workers.
- ▶ Waiters.
- ▶ Cleaners.
- ▶ Painters.
- ▶ Dental workers.
- ▶ Laboratory technicians.

(British Thoracic Society and Scottish Intercollegiate Guidelines Network 2011)

Spirometry should only be carried out by practitioners who have been trained in the task. Poor spirometry technique and incorrect interpretation can result in incorrect diagnosis and potentially inappropriate treatment (Bostock-Cox 2010).

Peak expiratory flow measurement

PEF is a simple test of lung function that measures the maximum flow of air achievable from a forced expiration that is started from a position of maximum lung inflation (Booker 2007). PEF measurement is easy and quick to explain and once the patient can perform the test well, it can have an important role in monitoring patients with asthma. The correct way to take a peak flow reading is explained in Box 3.

Predicted lung function values from either spirometry or PEF are related to a person's age, gender, height and ethnicity (Booker 2007, 2008).

Although both PEF and spirometry can be used in diagnosis, it is important to remember that results may be normal if the measurement is taken when there is no underlying obstruction of the airways (Scullion 2005). PEF can be a useful test in assessing variability, which is one of the key features of asthma (Bostock-Cox 2010). Assessing the extent of variability in PEF readings requires the patient to keep a peak flow diary for at least two weeks. PEF should be measured in the morning and at night, and any medications, symptoms or factors that may be relevant to triggering symptoms should be documented (Scullion 2005). In people with asthma, PEF is often lower in the morning – the so called early morning dips – and higher in the evening (Bostock-Cox 2010). The normal daily variations in airway size are exaggerated in patients with asthma. Significant variability is present if peak flow readings vary by 20% or more, especially if there are morning dips. PEF is best used to provide an estimate of variability of airflow from multiple measurements made over a period of at least two weeks (BTS and SIGN 2011).

Spirometry and PEF are also important in the measurement of reversibility. Reversibility refers to the difference in lung function before and after treatment (Bostock-Cox 2010). For asthma diagnosis, spirometry is usually assessed before and after the administration of an inhaled short-acting beta-agonist, for example salbutamol; responsiveness of FEV₁ by 15% or 200mL (whichever is the greater)

is indicative of asthma. An increase of more than 15% and at least 60L/min of PEF, 15-20 minutes after the inhalation of a short-acting beta-agonist, is indicative of asthma (Holgate and Douglass 2010).

Complete time out activity 3

Differential diagnosis

Symptoms associated with asthma such as cough, wheeze and breathlessness can also arise from other causes, so differential diagnosis should be considered. Other disorders that can present with symptoms similar to asthma are listed in Box 4.

In addition to obtaining a detailed medical history and conducting lung function tests, careful physical examination can help with differential diagnosis (Haldar and Pavord 2008, Sarver and Murphy 2009). Clinical examination is generally normal in individuals with well-controlled asthma symptoms. Those with persistent symptoms may have features of obstructive airway disease such as chest hyperexpansion and expiratory wheeze. The BTS and SIGN (2011) guideline on the management of asthma in adults refers to the concept of probability in making a diagnosis of asthma.

BOX 3

Correct way to take a peak flow reading

- ▶ The patient should be standing if possible.
- ▶ The patient should be shown how to hold the peak flow meter horizontally to the mouth, making sure the fingers are free of the cursor.
- ▶ The patient should be taught to take the deepest inhalation possible, then to seal the lips around the mouthpiece and blow out as hard as possible.

(Scullion 2005)

BOX 4

Examples of differential diagnoses

- ▶ Chronic obstructive pulmonary disease.
- ▶ Ischaemic cardiac pain.
- ▶ Heart failure.
- ▶ Gastroesophageal reflux disease.
- ▶ Inhaled foreign bodies and upper airway obstruction.
- ▶ Vocal cord dysfunction.
- ▶ Hyperventilation.

(Sarver and Murphy 2009, Rees 2010)

The practitioner has three options after initial assessment and examination:

- ▶ High probability – diagnosis of asthma likely. For adults in this category the guideline recommends a trial of treatment based on a stepwise approach. Treatment should continue if the response is positive, but in the absence of a response further investigation or referral is recommended.
- ▶ Intermediate probability – diagnosis uncertain. For adults in this group the guideline recommends a trial of treatment if the FEV₁/FVC is less than 70%. Adults with an FEV₁/FVC more than 70% require further investigation or referral.
- ▶ Low probability – other diagnosis likely. For adults in this category investigation or treatment for whatever condition is suspected should be initiated and referral considered if appropriate.

Complete time out activity ④

Pharmacological treatment

The BTS and SIGN (2011) stepwise approach to asthma management (Table 1) helps guide the practitioner when prescribing treatment. The stepwise approach provides advice on drug classes and suitable doses. Patients should start treatment at the step most appropriate to the initial severity of their asthma so that early control is achieved. Treatment should be stepped up when required and stepped down when control is good. Practitioners should always revisit the diagnosis and check adherence to existing medication and inhaler technique before stepping up treatment (BTS and SIGN 2011).

Short-acting beta₂ agonists

First-line treatment for mild intermittent asthma is a short-acting beta₂ agonist, either salbutamol or terbutaline, taken by inhalation (Barnes 2008, Rees 2010). Short-acting beta₂ agonists work on the beta₂ adrenergic receptors in the smooth muscle of bronchial tissue, producing bronchodilation and relieving the symptoms of chest tightness and breathlessness (Scullion and Holmes 2010).

Inhaled corticosteroids

If symptom control is not achieved with a bronchodilator, an inhaled corticosteroid should be added for adults who need to use an inhaled beta₂ agonist three times a week or more, who are symptomatic three times a week or more or are waking one night a week (BTS and SIGN 2011). Inhaled corticosteroids

such as beclometasone dipropionate, budesonide and fluticasone propionate, are the most effective preventive therapy for asthma (Rees 2010). These agents reduce inflammation and decrease airway hyper-responsiveness and the frequency of exacerbations (Holgate and Douglass 2010).

Although the above-mentioned drugs appear to be equally effective (British National Formulary (BNF) 2011) there are differences in potency between beclometasone dipropionate, budesonide and fluticasone propionate, which means that doses are not equivalent. The recommended doses of inhaled corticosteroids are set out in the BTS and SIGN (2011) guideline.

Long-acting beta₂ agonists

A long-acting beta₂ agonist can also be added as part of the stepwise approach if, as required, use of a short-acting beta₂ agonist and regular use of an inhaled corticosteroids do not control symptoms. Long-acting beta₂ agonists include salmeterol and formoterol. These agents act by relaxing the airways, enhancing mucociliary clearance and decreasing vascular permeability (Holgate and Douglass 2010). However, the long-acting beta₂ agonist should be discontinued if there is no response and the dose of inhaled corticosteroid increased (BTS and SIGN 2011). Long-acting beta₂ agonists should not be used as monotherapy and should only be started in patients who are on inhaled corticosteroids, and the inhaled corticosteroid should be continued (BTS and SIGN 2011).

Inhaled long-acting beta₂ agonists and inhaled corticosteroids can be given in a combination inhaler rather than separate inhalers once effectiveness is established. Combination inhalers are considered to aid compliance and have the advantage of guaranteeing that the long-acting beta₂ agonist is not taken without the inhaled steroid (BTS and SIGN 2011). Examples of combination inhalers include Seretide and Symbicort (BNF 2011).

Leukotriene receptor antagonists

If there is persistent poor control of asthma, the addition of a fourth drug such as a leukotriene receptor antagonist is recommended. Leukotriene receptor antagonists, such as montelukast and zafirlukast, affect the production and/or activity of leukotrienes, which are responsible for airway inflammation and hyper-responsiveness, mucus production, oedema and bronchoconstriction



③ Having read the previous section, define spirometry PEF and summarise the advantages and disadvantages of each in asthma diagnosis. Access this link, which will take you to the BTS and SIGN (2011) guideline on the management of asthma: www.sign.ac.uk/pdf/qrg101.pdf Scroll down to pages four and five and read the guidance on diagnosis in adults.

④ Access the BTS and SIGN (2011) guideline on the management of asthma. Read pages eight and nine on pharmacological management from the quick reference guide. A patient with asthma is taking an inhaled short-acting beta-agonist as required and 400mcg of inhaled corticosteroid. However, the patient is still experiencing symptoms. What does the BTS and SIGN (2011) guideline suggest should happen next?

(Barnes 2008, Kuebler *et al* 2008, Rees 2010). Overall, the effects achieved with these drugs are less than those achieved with inhaled corticosteroids (Rees 2010). The leukotrienes can be effective in patients with aspirin or exercise-induced asthma (Barnes 2008). Generally, patients seem to differ in their response to leukotriene inhibitors and it is difficult to predict which patients will respond best (Barnes 2008).

Methylxanthines

The methylxanthine theophylline is an effective bronchodilator that may also have anti-inflammatory properties (Rees 2010). However, the drug has a narrow therapeutic index which means the toxic dose is only a little higher than the effective dose and toxic

levels can cause fatal side effects (Anwar 2008). Appropriate dosing is essential and monitoring is advisable when treatment is started, and at regular intervals thereafter (Holgate and Douglass 2010).

Before introducing drugs such as theophylline it is important to revisit the diagnosis and check compliance with treatment and inhaler technique.

Complete time out activity 5

Oral corticosteroids

Some patients with very severe asthma that is not controlled with high-dose inhaled corticosteroids, and who have also been tried on a long-acting beta agonist, a leukotriene antagonist or a theophylline, will require regular long-term oral corticosteroid tablets

TABLE 1

Drugs used to treat asthma, their mechanism of action and their place in the stepwise approach to asthma management

Step	Medications	Mechanism of action
Step 1 Mild intermittent asthma	Inhaled short-acting beta ₂ agonist as required, for example salbutamol or terbutaline.	Bronchodilation.
Step 2 Regular preventer therapy	Add inhaled corticosteroid 200-800mcg per day. Start at dose of inhaled corticosteroid appropriate to severity of the disease.	Reduces inflammation and decreases airway hyper-responsiveness.
Step 3 Initial add on therapy	Add inhaled long-acting beta ₂ agonist (LABA), for example salmeterol or formoterol.	Bronchodilation, enhances mucociliary clearance, decreases vascular permeability.
	Assess response and continue LABA if response is good.	
	Benefit from LABA, but control still inadequate, continue LABA, but increase inhaled corticosteroid to 800mcg per day (if not already on this dose).	
	If control is still inadequate, introduce trial of other therapies, for example, a leukotriene receptor antagonist or slow release theophylline.	
Step 4 Persistent poor control	Consider trials of increasing inhaled corticosteroid up to 2mg per day.	
	Addition of a fourth drug: Leukotriene receptor antagonist or Slow release theophylline	Reduces airway inflammation and hyper-responsiveness, mucus production, oedema and bronchoconstriction. Bronchodilation. Anti-inflammatory properties.
Step 5 Continuous or frequent use of oral corticosteroids	Use daily oral corticosteroid in lowest dose providing adequate control. Maintain high dose inhaled corticosteroid at 2mg per day. Consider other treatments to minimise the use of oral corticosteroid. Refer patient for specialist care.	

(Rees 2010, British Thoracic Society and Scottish Intercollegiate Guidelines Network 2011)

(BTS and SIGN 2011). However, oral corticosteroid tablets used long-term are associated with a risk of systemic side effects and should be used in the lowest dose providing adequate control. Before proceeding to oral corticosteroids, patients with inadequately controlled asthma should be referred to specialist care (BTS and SIGN 2011).

Inhaler devices and technique, and adherence to treatment

While the BTS and SIGN (2011) guideline gives advice on drug classes and possible doses, it does not provide information on the choice of device for inhaled therapies. The inhaler device used to deliver the drugs is as important as the drugs themselves (Bostock-Cox 2010). An appropriate inhaler device and correct inhaler technique are important for adherence to treatment and control of symptoms. A variety of devices is available for the administration of inhaled therapies, including spacers, pressurised meter dose inhalers and dry powder devices. Each type of device has advantages and disadvantages. For example metered dose inhalers are quick to use and easy to transport, but good co-ordination is needed to depress the canister and inhale at the same time. This can be difficult for some patients, including children. Dry powder inhalers are easier to use because they are breath actuated, but may require a high inspiratory flow for optimum use (Madoc-Sutton *et al* 2009). However, most patients can learn how to use an inhaler unless they have significant cognitive impairment, and any of the devices can work well if the patient can use them (Madoc-Sutton *et al* 2009). It is important that healthcare professionals caring for patients with asthma discuss preferences for inhaler devices. The choice of device may be determined by the choice of drug and different patients are likely to have different requirements, therefore no single device will satisfy the needs of every patient (Madoc-Sutton *et al* 2009). If a patient cannot use a particular device another should be tried.

The BTS and SIGN (2011) guideline emphasises that inhalers should only be prescribed after patients have received training in the use of the device and have demonstrated satisfactory technique. It is essential to check technique regularly.

The BTS and SIGN guideline (2011) no longer recommends nebuliser therapy for most asthma care. In some situations, nebulisers can be detrimental, particularly if patients use them

to treat the symptoms (with bronchodilators) rather than treating the underlying cause (inflammation) with corticosteroids. The clinical effectiveness of nebuliser therapy is no more efficient than using a metered dose inhaler and a spacer, and the latter is more cost-effective and convenient (Kelly and Lynes 2011). In acute asthma, BTS and SIGN (2011) recommend the delivery of high-dose bronchodilators via a metered dose inhaler and spacer as first-line management in all but life-threatening emergencies.

Complete time out activity 6

Supported self-management

Key components of an optimal management strategy include partnership working between healthcare professionals and patients, comprehensive patient and caregiver information, appropriate medication and personalised asthma action plans (Sarver and Murphy 2009). Partnership working and regular monitoring provide the opportunity to elicit patient and carer perceptions of the disease and its treatment. Many misperceptions can exist (Cornforth 2010) therefore partnership working is important to address individual concerns and give information that can help patients improve their understanding of the disease.

There is a common perception among patients and carers that there are numerous environmental and dietary triggers of asthma, and that avoidance of these will reduce the need for drug therapy. The BTS and SIGN (2011) guideline highlights studies that explore interventions introduced before the onset of asthma to reduce incidence of the condition, as well as interventions introduced after the onset of asthma to reduce its effect. Such interventions include aeroallergen (such as pollen or spores), food allergen and house dust mite avoidance. However, evidence that such non-pharmacological interventions are effective is difficult to obtain and better controlled intervention studies are required.

Partnership working and regular monitoring enables the practitioner to record and monitor peak flow. In some individuals, changes can occur in PEF before the onset of acute symptoms. Early detection of an exacerbation can allow appropriate treatment to be given. PEF readings can also demonstrate the severity of asthma when compared with previous readings enabling the instigation of a self-management plan (Douglass and Holgate 2010). Portable meters for the measurement



5 Make a list of the inhaler devices that you are aware of. List the factors that you think are important about inhaler devices.

6 Access the following link and look at the different inhaler devices: www.asthma.org.uk/health_professionals/interactive_inhaler_demo Are there any devices with which you are unfamiliar? Watch the inhaler technique demonstration. Aside from drug therapy, list any other factors that you think are important in the management of asthma.



7 Access the following link to view a sample asthma action plan: <http://tinyurl.com/yfahlp>

8 Now that you have completed the article, you might like to write a practice profile. Guidelines to help you are on page 60

of PEF can be obtained on NHS prescription (Booker 2007, Holgate and Douglass 2010).

Partnership working and regular monitoring is also an opportunity to check and correct inhaler technique and discuss the use of medication. A partnership in which there is open dialogue is most influential in medication adherence, which is central to good symptom control (Sarver and Murphy 2009).

The importance of self-management should be emphasised for all patients with asthma. It is essential they are aware of the signs and symptoms of poor control and know how to monitor their symptoms. Self-management plans are individualised written protocols that set down specific courses of action on the basis of asthma symptoms and/or peak flow recordings. Action plans that contain appropriate information have been shown to improve asthma outcomes in terms of improving morbidity and reducing the frequency of hospitalisations (Halder and Pavord 2008). **Complete time out activity 7**

Conclusion

Asthma is a disease of the airways, characterised by inflammation and associated with airway hyper-responsiveness, which can result in episodes of breathlessness, chest tightness, wheezing and cough. Genetic factors, environmental influences and specific trigger factors are implicated in the development of the disease.

A thorough clinical history and objective measurement of lung function are important in establishing a reasonably certain diagnosis of asthma. A range of drugs is used in the management of the disease, and clinical guidelines advocate a stepwise approach to drug therapy, where treatment is stepped up when required and stepped down when control is good. Alongside pharmacological management of the disease, partnership working between patients and healthcare professionals, personalised written asthma action plans, information and education are central to improving the quality of life of patients with asthma **NS** **Complete time out activity 8**

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